



Data Analysis and Integration

Project

In this project you will build an analytical data warehouse to investigate **Portugal's Economic and Human Development: A Century of Transformation**.

The challenge is to integrate two international datasets and an historical events dataset into a coherent star schema and use OLAP technologies to analyze how economic growth, structural transformation and wellbeing evolved through the last century.

The source datasets encode the observation and/or estimates of economic and societal indicators per year for distinct countries.

Data sources

The data sources are:

1. [Maddison Project Database 2023](#) (**maddison_indicators.csv**) → historical reconstruction from 1900 to 2022 of long-run GDP per capita, GDP and population.
2. [World Bank World Development Indicators \(WDI\)](#) (modern **wdi_indicators.csv**) → modern economic, structural and social indicators (GDP growth, GNI per capita, sector value added, employment by sector, life expectancy, urbanization, trade).

Note: Most social series start circa 1960. Sector value added often starts in the 1990s (Portugal and most comparison countries from 1995). Some transition-economy series also start in the 1990s.

3. Historical Events Dataset (**historical_events.csv**) → with 387 major economic, political, infrastructure and social events for Portugal and the comparison countries (1900-2025).

These sources have different temporal coverage and formats. The provided files already share ISO3 codes (country_iso3) and the same country names; you still need a unified mapping for missing entities (the EU aggregate is in WDI and events only), historical breaks, and country groups. Managing these differences is an explicit learning objective.

The analysis focuses on the following countries:

- Focus country: Portugal.
- Western European comparison: Spain, France, Germany, Belgium, Italy, Ireland, Greece.
- Central/Eastern European comparison: Poland, Czechia, Hungary, Romania.
- Post-Soviet comparison: Estonia, Latvia, Lithuania.
- European Union aggregate (EUU), where available.

The data warehouse should use a star schema with three typed fact tables sharing two conformed dimensions, plus a sector dimension:

- FACT_ECONOMY with one row per country per year: measures gdp_per_capita and gdp from Maddison; gdp_growth_pct, exports_pct_gdp and imports_pct_gdp from WDI.

- FACT_SOCIETY with one row per country per year: measures population from Maddison; gni_per_capita, life_expectancy and urban_pop_pct from WDI.
- FACT_SECTOR with one row per country per year per sector (from WDI): measures value_added_pct_gdp, value_added_usd, employment_pct.
- DIM_COUNTRY; with surrogate key, ISO3 code, name and country group.
- DIM_TIME; with surrogate key, year and decade.
- DIM_SECTOR; with surrogate key, sector name and broad sector (hierarchy: Primary/Secondary/Tertiary → Agriculture/Industry & Manufacturing/Services).

Note: For example “manufacturing” (NV.IND.MANF) is part of industry (NV.IND.TOTL) in the national accounts; do not add the two. Employment shares exist only for agriculture, industry and services; employment_pct is empty for Manufacturing (there is no SL.MANF series). The EU aggregate (EUU) has WDI and events but no Maddison GDP or population, so those FACT_ECONOMY and FACT_SOCIETY measures stay empty for EUU. Irish GDP is distorted by multinational activity; GNI in FACT_SOCIETY is the better income comparator for Ireland.

Tasks

For the project execute the following tasks:

1. Create a staging database and profile data. Write SQL scripts to create a staging table schema with precise data types. Then load the provided files maddison_indicators.csv, wdi_indicators.csv and historical_events.csv. After loading, run exploratory SQL queries to list distinct countries per source, inspect year ranges, verify data integrity, and support the temporal and geographic coverage analysis in the data dictionaries created later in this question.

For each data source, create a data dictionary documenting, where applicable:

- column names, types, and descriptions;
 - units of measurement, row counts, number of entities, and year span;
 - temporal and geographic coverage, including missing values and relevant coverage gaps; *this may be described by measure or by country, depending on what is most appropriate for each dataset*;
 - relevant patterns or particularities in the data.
2. Country-code reconciliation. The provided files already use ISO3 codes (country_iso3) and aligned country names, but you still need a unified mapping: the EU aggregate is in WDI and events only, and each country needs a country_group.

Using Apache Hop, build a pipeline that:

- Extracts the distinct country names and codes from each source.
- Produces a unified country mapping table (**iso3_code, country_name, country_group**).
- Loads the final mapping into a CSV (.csv) output file;
- Loads the same mapping into the staging database created in the previous question.

The following country groups are defined for the project:

Country Group	Countries
Focus	Portugal
Western Europe	Spain, France, Germany, Belgium, Italy, Ireland, Greece
Central/Eastern Europe	Poland, Czechia, Hungary, Romania
Post-Soviet	Estonia, Latvia, Lithuania
Benchmark	European Union

3. Design the data warehouse. The schema should include the three fact tables (**FACT_ECONOMY**, **FACT_SOCIETY**, **FACT_SECTOR**), the three dimension tables (**DIM_COUNTRY**, **DIM_TIME**, **DIM_SECTOR**), and your chosen integration approach for the historical events data.

You must decide how to integrate the historical events dataset into the schema. Three valid approaches exist:

- 1) As a standalone reference table joined via year.
- 2) As a bridge table linking to DIM_TIME and DIM_COUNTRY.
- 3) A new DIM_EVENT dimension.

You must justify your choice.

The data warehouse design must also:

- Use surrogate keys for all dimensions
- Produce an ER diagram showing the complete schema.
- Write a short and compact justification explaining why three separate fact tables (different grains and subject areas); why DIM_SECTOR only applies to FACT_SECTOR; and how you chose to integrate the events data.

Notes: Write also an SQL script to create a star schema as a separate database.

4. Develop the ETL pipelines to populate the data warehouse from the staging database. Using Apache Hop, create separate pipelines for each dimension and fact table, and create a workflow to execute all pipelines in the correct order.

The suggested pipelines are the following:

- DIM_COUNTRY (incorporating the country mapping from Task 2).
- DIM_TIME (generate year rows from the minimum to maximum year across sources; compute the decade).
- DIM_SECTOR (populate from a predefined list of sectors with the **broad_sector** hierarchy).
- FACT_ECONOMY (Maddison: gdppc and gdp, plus WDI: growth and trade; pivot from long format; surrogate key lookups).
- FACT_SOCIETY (Maddison: population, plus WDI: GNI, life expectancy and urbanization; pivot from long format; surrogate key lookups).
- FACT_SECTOR (WDI: sector data; pivot and map to DIM_SECTOR; surrogate key lookups).
- The Events table or dimension (as designed in Task 3).

Note: For each pipeline, present a screenshot of the full pipeline, screenshots of the key transform configurations, and evidence of the resulting row counts.

5. Cube definition. Using Pentaho Schema Workbench (PSW), create an XML cube definition. Configure Pentaho Server and, using Saiku Analytics, perform at least three multidimensional analyses. The cube should include:
 - Country hierarchy: Country Group → Country.
 - Time hierarchy: Decade → Year.
 - Sector hierarchy: Broad Sector → Sector.
 - Measures from all three fact tables.

Required Saiku analyses:

- A. Long-run growth; analyze GDP per capita for Portugal and comparison countries since 1900, by year and decade. Annotate key events that explain turning points.
- B. European comparison; analyze GDP, GNI, Life expectancy of Portugal versus Western European countries since 1975 and versus Eastern European countries from 1990 onward.
- C. Structural transformation; analyze agriculture, industry, manufacturing and services as a share of added pct GDP, using the sector hierarchy drill-down.

Note: Present the PSW schema XML, a screenshot of the tree structure, screenshots of each Saiku analysis, and export at least one result to Excel (XLS/XLSX). Use the events data to explain observed patterns in each analysis.

6. Write MDX queries that use calculated members and the cube hierarchies (not only a Saiku drag-and-drop).
 - A. **Query 1:** Using a calculated member and a tuple, show Portugal's GDP per capita as a ratio of Spain's GDP per capita, by decade from 1960 to 2020. Use the time hierarchy (Decade). This answers whether Portugal converged with or diverged from Spain.
 - B. **Query 2:** Using the sector hierarchy (Children or Descendants), show Portugal's value_added_pct_gdp at both Broad Sector and Sector level. Put 1995 and 2020 on the columns as tuples. Sector shares start in 1995 for Portugal and most comparison countries.

Note: Present the MDX queries and screenshots of the results.

7. Answer the following questions (answer should be brief):
 - A. How did Portugal's GDP per capita evolve since 1900? Which decades recorded the highest growth (you may compute decade change from Maddison gdppc; WDI GDP growth is annual and mostly from 1961, or 1991 in the East)? What historical events explain the major turning points?
 - B. How does Portugal compare with Spain, France, Germany, Belgium, Italy, Ireland and Greece across the 20th and 21st centuries?
 - C. Did Portugal converge with or diverge from Western Europe? When were the strongest convergence periods? (e.g., EU accession in 1986; euro accounting adoption in 1999; notes and coins in 2002)

- D. How do Poland, Czechia, Hungary, Romania, Estonia, Latvia and Lithuania compare with Portugal after their post-communist transitions (1990 onward)?

Note: You may solve the questions either using MDX queries or Saiku analyses. In both cases, submit the corresponding queries/analyses and screenshots of the results. For each question, also include a brief written analysis summarizing the main findings and complementing the presented results.

8. Create the final report and data quality discussion:

- A. Report: Using PRD, produce a report titled **Portugal's Economic and Human Development: A Century of Transformation**. The report should include at least one chart (e.g., GDP per capita over time by country) and at least one table with grouped data. Where relevant, annotate with historical events.
- B. Discussion (1 page): Discuss the main data-quality issues encountered: country-name and coverage differences between Maddison and WDI; temporal gaps (Maddison from 1900 to 2022, WDI mostly from ~1960, with later starts for sector and transition-economy series); missing values for specific countries and years; the WDI long-to-wide pivoting challenge; empty Maddison measures for the EU aggregate. Also discuss the ETL bottlenecks you encountered and the most interesting patterns you discovered in the data.

Note: Present screenshots of the report in design and preview mode. Export the report in PDF format. Save the report as .prpt.

Submission

The project should be delivered in a zip file named **group_XX.zip** (where **XX** is a two-digit group number) and submitted via Fénix until 23:59 of 21/10/2026.

At the root of the zip file, there should be a slide presentation (**prepared with PowerPoint or similar**) in PDF format, named **slides_XX.pdf** (where **XX** is a two-digit group number).

The slides should present the outputs (code, relevant screenshots, results, discussion, etc.) for each requested task. You may organize the slides to make it easier to locate a specific task.

In addition, the zip file should contain a folder for each task (**task_01/**, **task_02/**, ..., **task_08/**) with each folder containing the files produced for that task:

```
group_XX.zip
  slides_XX.pdf
  task_01/
    create_staging.sql
    load_staging.hpl
    exploratory_queries.sql
    data_dictionary_maddison.pdf
    data_dictionary_wdi.pdf
    data_dictionary_events.pdf
  task_02/
    country_matching.hpl
    country_mapping.csv
```

task_03/
 create_dw.sql
 schema_diagram.png (or .pdf)
 design_justification.pdf
task_04/
 dim_country.hpl
 dim_time.hpl
 dim_sector.hpl
 fact_economy.hpl
 fact_society.hpl
 fact_sector.hpl
 events.hpl
task_05/
 schema.xml
 analysis_1_growth.xlsx
 analysis_2_comparison.xlsx
 analysis_3_transformation.xlsx
 events_explanation.pdf
task_06/
 mdx_queries.txt (or .sql)
task_07/
 analysis_a.xlsx or mdx_a.txt (or .sql)
 analysis_b.xlsx or mdx_b.txt (or .sql)
 analysis_c.xlsx or mdx_c.txt (or .sql)
 analysis_d.xlsx or mdx_d.txt (or .sql)
 questions_explanation.pdf
task_08/
 report.prpt
 report_preview.pdf

Note: Please follow the submission instructions. Evaluation elements that are not found as prescribed above will not be considered for grading purposes. Also, please do not wait until the last minute to submit, as late submissions will not be accepted.

Appendix A. Maddison source database units reference

Indicator Code	Description	Units	Fact table
gdppc	GDP per capita	2011 international dollars per person (PPP, constant 2011 prices). An international dollar is a PPP dollar: one 2011 int-\$ buys what one US dollar bought in the United States in 2011. That is how we can compare countries across years.	FACT_ECONOMY (gdp_per_capita)
pop	Population (mid-year)	Thousands of people (mid-year). Portugal 1900 = 5404 means 5.404 million	FACT_SOCIETY (population)
gdp	Gross domestic product (total)	2011 international dollars (total GDP). Computed as $gdppc \times pop \times 1000$	FACT_ECONOMY (gdp)

Appendix B. WDI source database units reference

FACT_ECONOMY indicators (WDI):

Indicator Code	Description	Units	Fact table
NY.GDP.MKTP.KD.ZG	GDP growth (annual %)	Annual percentage change of real GDP	FACT_ECONOMY (gdp_growth_pct)
NE.EXP.GNFS.ZS	Exports of goods and services (% of GDP)	Percentage of GDP	FACT_ECONOMY (exports_pct_gdp)
NE.IMP.GNFS.ZS	Imports of goods and services (% of GDP)	Percentage of GDP	FACT_ECONOMY (imports_pct_gdp)

FACT_SOCIETY indicators (WDI):

Indicator Code	Description	Units	Fact table
NY.GNP.PCAP.CD	GNI per capita, Atlas method (current US\$)	Current US\$ (Atlas method)	FACT_SOCIETY (gni_per_capita)
SP.DYN.LE00.IN	Life expectancy at birth, total (years)	Years	FACT_SOCIETY (life_expectancy)
SP.URB.TOTL.IN.ZS	Urban population (% of total population)	Percentage of total population	FACT_SOCIETY (urban_pop_pct)

FACT_SECTOR indicators (WDI):

Indicator Code	Description	Units	Fact table
NV.AGR.TOTL.ZS	Agriculture, forestry, and fishing, value added (% of GDP)	Percentage of GDP	FACT_SECTOR (value_added_pct_gdp)
NV.AGR.TOTL.CD	Agriculture, forestry, and fishing, value added (current US\$)	Current US\$	FACT_SECTOR (value_added_usd)

NV.IND.TOTL.ZS	Industry (including construction), value added (% of GDP)	Percentage of GDP	FACT_SECTOR (value_added_pct_gdp)
NV.IND.TOTL.CD	Industry (including construction), value added (current US\$)	Current US\$	FACT_SECTOR (value_added_usd)
NV.IND.MANF.ZS	Manufacturing, value added (% of GDP)	Percentage of GDP	FACT_SECTOR (value_added_pct_gdp)
NV.IND.MANF.CD	Manufacturing, value added (current US\$)	Current US\$	FACT_SECTOR (value_added_usd)
NV.SRV.TOTL.ZS	Services, value added (% of GDP)	Percentage of GDP	FACT_SECTOR (value_added_pct_gdp)
NV.SRV.TOTL.CD	Services, value added (current US\$)	Current US\$	FACT_SECTOR (value_added_usd)
SL.AGR.EMPL.ZS	Employment in agriculture estimate (% of total employment)	Percentage of total employment	FACT_SECTOR (employment_pct)
SL.IND.EMPL.ZS	Employment in industry estimate (% of total employment)	Percentage of total employment	FACT_SECTOR (employment_pct)
SL.SRV.EMPL.ZS	Employment in services estimate (% of total employment)	Percentage of total employment	FACT_SECTOR (employment_pct)